AAROHAN 2026

Ascent of Knowledge

XI

Sample Paper

Name

Roll No.

BITS Pilani
Pilani Campus

APOGEE 2026

Instructions: -

- Read the passage carefully, it contains all the necessary details for solving the questions that follow it.

- The questions will be based on Physics, Chemistry, Mathematics and Analytical Skills.

- The questions can test your understanding of one or more of the above subjects at once.

- Questions marked with an asterisk(*) **may** have more than one option as the correct answer(s). You must select all correct option(s) to earn marks.

- **All other questions have only one correct alternative**. The marking scheme is
  - **+3 -> Correct Answer**
  - **0-> Unattempted Question**
  - **-1 -> Incorrect Answer**

- For **multiple correct answer** type questions the marking is scheme is
  - **+4 -> All correct options selected**
  - **0 -> All other cases**

- If an assumption is to be made, it will be mentioned in the question itself, along with all other necessary data/information required. Try to crack with the information that has been given- we test your **thinking process** not rote knowledge.

- Before you start this exam, **calm down** and remind yourself that **you can crack this! Keep moving forward, do not get stuck on one question.**

- **Best of Luck!!**

MURDER

The deadly gas phosgene is a highly toxic, colourless gas at room temperature and standard pressure that condenses at 0°C to a fuming liquid. Its molecular formula is $COCl_2$.
Phosgene at a concentration of 1 ppm in the air causes little or no immediate irritation, but after a latent period of some hours, it may cause severe pulmonary edema and at concentrations of 4-10 ppm in air can cause irritation of the respiratory tract and eyes. Lethal dose of phosgene in humans is approximately 500 ppm/min of exposure or exposure at 3 ppm for 170 min is equally as fatal as exposure at 30 ppm for 17 min. Exposure occurs by inhalation and the fact that phosgene is only a slightly water-soluble gas and that due to this, significant irritation of upper respiratory tract and eyes may not occur, leading to prolonged exposure. Phosgene exerts its toxicity through the acylation of proteins as well as through the release of hydrochloric acid. The amino, hydroxyl and sulfhydryl groups in proteins appear to be the target for acylation, leading to marked inhibition of several enzymes related to energy metabolism and a breakdown of the blood-air barrier.
In the terminal phase, phosgene toxicity leads to the accumulation of fluid in the lungs. The severity of the edema(swelling) increases potentially, resulting in decreased gas exchange as the fluid gradually rises from the alveoli to the proximal segments of the respiratory tract. Agitated respiration may cause the protein-rich fluid to take a frothy consistency. A severe edema may result in an increased concentration of hemoglobin in the blood and congestion of the alveolar capillaries.

POWER OF DNA:

DNA fingerprinting includes a classifications system to find a number for fingerprints. Fingerprints can have different patterns namely loop, whorl and arch.
It consists of five fractions, in which R stands for right, L for left, i for index finger, m for middle finger, t for thumb, r for ring finger and p (pinky) for little finger. The fractions are as follows:

$$\frac{R_i}{R_t} + \frac{R_r}{R_m} + \frac{L_t}{R_p} + \frac{L_m}{L_i} + \frac{L_p}{L_r}$$

The numbers assigned to each print are based on whether or not they are whorls. A whorl in the first fraction is given a 16, the second an 8, the third a 4, the fourth a 2, and 0 to the last fraction. Arches and loops are assigned values of 0. Lastly, the numbers in the numerator and denominator are added up, using the scheme:

$$\frac{(R_i + R_r + L_t + L_m + L_p)}{(R_t + R_m + R_p + L_i + L_r)}$$

1 is added to both top and bottom, to exclude any possibility of division by zero. For example, if the right middle finger and the left index finger have whorls, the fraction used is:

0/0 + 0/8 + 0/0 + 0/2 + 1/1

The resulting calculation is:
(0 + 0 + 0 + 0 + 1) / (0 + 8 + 0 + 2 + 1) = 1/11
Another technique used to investigate murders is DNA profiling. A DNA sample taken from a person is compared with the evidence found at the crime scene. The process of DNA testing has 4 main steps.
- Amplification: A technique called polymerase chain reaction (PCR) is used in this step with the help of which millions of copies of the DNA are generated.
- Capillary Electrophoresis: In this step, various DNA molecules are separated from each other with the help of electric current.
- Extraction: In this step, the nucleus is broken open and the DNA molecules are released into the solution.
- Quantitation: In this step, the quantity and quality of the DNA present in a the sample is measured and assessed.

THE PSYCHOTIC MANIAC

You might have heard of the word “Serial Killer” but a lunatic murderer seems to be a “Sequence Killer” for his mad obsession with killing people every day from a different village, in a way that the number of people killed in each consecutive village follows a sequence such as 3, 4, 7, 14, 27, 48 and so on. He has decided to start his streak of murders in a sequence of villages that have populations in an AP, starting with 100 people in the 1st village then 150 in the 2nd, and so on. Our “Sequence Killer” starts with killing 3 people in the 1st village, then killing 4 people in the 2nd village and continues this process till the number of murders he wishes to perform is more than the population of the village in the sequence.

QUESTIONS:-

Q1. A serial killer plans on killing a family by poisoning the ventilation ducts of their house with phosgene. Which of the following substances could be used to treat or at least minimize the effect of phosgene in the victims?
(a) HCl
(b) Ammonia
(c) Baking soda
(d) Water

Q2. Suppose a victim is in the terminal phase of phosgene poisoning. We know that hemoglobin plays the role of oxygen carrier through our blood system. Which of the following statements hold true for this phase?
(a) Immediate increase in iron will be beneficial for the patient
(b) There is decreased partial pressure of oxygen
(c) Oxygenation of blood is poor
(d) Immediate increase in iron will be harmful for the patient

Q3. Suppose the killer releases phosgene into the house, where a family of 2(Phil and Amanda) lives. The amount inhaled is different in different rooms in the house. The probability of Phil dying is 0.4 and that of Amanda dying is 0.3. The probability of both of them dying is 0.36. What is the probability that at least one of them will die?
(a) 0.66
(b) 0.82
(c) 0.34
(d) 0.58

Q4. A murder has been reported and you are a forensic expert asked to investigate the crime scene. You collect the fingerprints and send them to the lab to run against the database. Prints of all fingers of both hands are procured and when you run it on the database, while there is a partial match found, there is a major power cut and you have to cross check the prints manually. The partial scan revealed that the number of scanned fingerprints was 3. There was a partial match with 4 suspects A, B, C, and D. Also it is known that A has whorls on right index and right thumb, B has whorls on right middle finger and right pinky, C has whorls on left thumb and right pinky and D has whorls on right ring finger, left middle finger, right pinky and left index finger. All the suspects may or may not have whorls on their other fingers. Which suspects can possibly be the murderer? (May be one or more)

(a) A
(b) B
(c) C
(d) D

Q5. Which of the following could be the correct order of the 4 stages in the process of DNA testing?
(a) Extraction, Amplification, Capillary Electrophoresis, Quantitation
(b) Extraction, Quantitation, Amplification, Capillary Electrophoresis
(c) Extraction, Amplification, Quantitation, Capillary Electrophoresis
(d) Extraction, Capillary Electrophoresis, Amplification, Quantitation

Q6. What would be the village wherein the psychotic maniac would not be able to commit his desired number of murders?

(a) 13th village
(b) 14th village
(c) 15th village
(d) 16th village

Q7. What would be the total number of murders(rounded off to the nearest thousands)performed by the psychotic maniac before he has to stop?
(a) 4 k
(b) 5 k
(c) 3 k
(d) 6 k

Q8. The three prime suspects of a murder are being questioned by the lawyer, the lawyer asks them: “Were all of you involved in the robbery?” The first suspect says: “I do not know.” The second suspect then says: “I do not know.” Finally, the third The suspect says: “No, not everyone was involved in the murder”. Assume you are the judge and it is known that none of the three suspects has lied. Then which suspects Would you be found guilty?

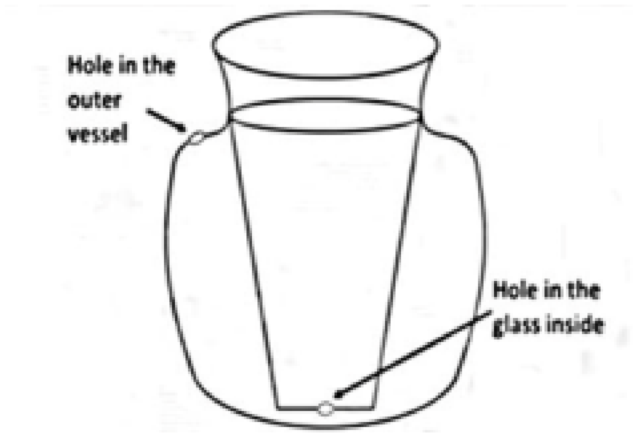
(a) 3rd suspect
(b) 2nd and 3rd suspect
(c) 1st and 2nd suspect
(d) 1st and 3rd suspect.

STREET MAGIC

Congratulations!!! You are the lucky winner of the contest of magic. You have been invited to attend the World's Greatest Magician competition. You have been given free access to roam the streets and see the various magic tricks shown by the finest magicians in the world. You would get a chance to interact and learn more about magic with the winner of our competition who would be crowned as the "World's Greatest Magician."
Magic many times has its roots in science. It uses basic concepts of science and sometimes even mathematics and builds up on them to make them appear as if supernatural forces are involved. The more the complexity of the thought behind it increases the more pleasing the magic trick becomes as it uses multiple components containing multiple concepts. Let us start with some of the easiest magic tricks that some of you must have seen in your life.

This magic trick of a man levitating while sitting has been one of the most popular tricks and it still confuses many people. This trick relies on the basics of physics. The location of the center of mass plays a vital role in this trick.

This is how the actual setup looks. Once the bottom support is covered with a cloth, and the hand is rested on one of the bars, it gives a feeling that a person is levitating in midair. Similarly, many magic tricks use this concept, where the setup has been hidden from viewers making them confused about how the magic works.
You now see a trick where one of the magicians keeps pouring water continuously after regular intervals from a glass even after emptying that glass. You are fortunate enough to see the diagram of the apparatus.



Sleight of Hand is also regarded as one of the most important aspects of performing magic. This type of magic does not rely on science much and can purely be mastered by practice. It does rely on the fact that our eyes can't usually see things that move in front of us in very less time.

This brings us to the next section of the fair where you see a lot of steam rising. One magician is performing a magic trick with a balloon. He blows a balloon using his air pump and keeps it on the table. After that he removes his kettle and starts pouring a steaming liquid on the balloon and to the surprise of everyone the balloon starts deflating. Once it has almost deflated, the magician places the balloon in normal water and the balloon starts inflating again to its full capacity.

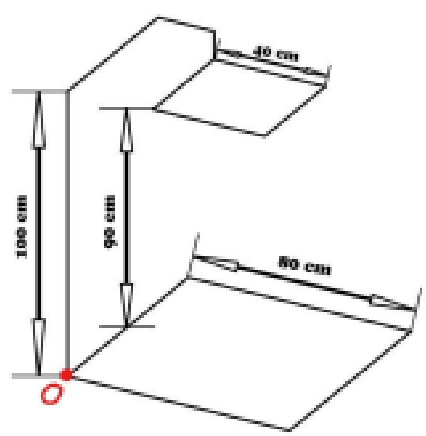
Now it is time for the championship all the magicians are here for. All the 20 magicians on the street have gathered to compete for the World Magic Championship. They have been split into two groups A and B and everyone from a group would be playing a match against each other. In a match, the two magicians competing against each other would be showing a magic trick one by one. Then the magician who has performed the best magic will be adjudged the winner of that match and will receive 5 points. The magic will be judged by the famous magicians, David Copperfield, Penn and Teller. After everyone in a group has played a match against each other, the top four teams from a group will proceed to the knockouts. In the knockouts, the player who wins all the matches is declared the winner (The first round in knockout will played in such a way that the nth magician in group A would play a match with (4-n)th magician in group B, and a similar pattern is followed for round 2, match 1's winner plays with match 3's winner and so on). A match cannot result in a tie but if the scores are tied, then the top 4 teams of a group are decided by the judge.

QUESTIONS:-

1. Suppose you are doing a magic trick with a coin of mass 4.5g in which you are trying to make it disappear from one hand by throwing it quickly with the help of your middle and ring finger, to your other hand. You throw it in such a way that you are barely able to catch it in your other hand. Find the force that you need to apply if the width of your hand is 80mm and distance between two hands is 30cm and the time of contact once the force is applied is 1 sec. (g=9.81 m/s)

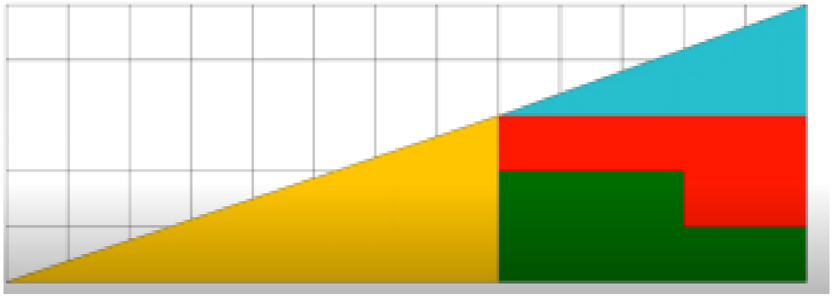
a. 0.034 N
b. 0.331 N
c. 0.015 N
d. 0.013 N

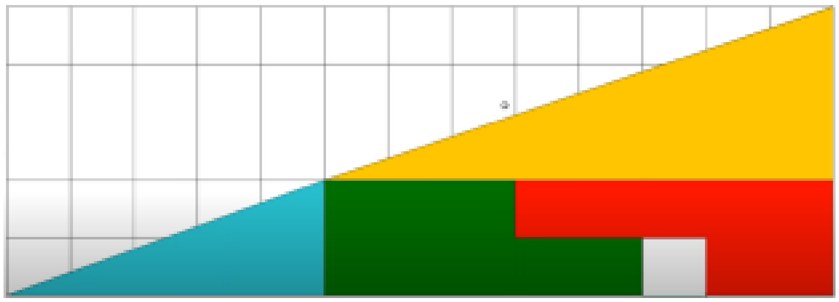
2. What can be the coordinate of the point where the normal force acts at, if the maximum mass that can be added on the below system in the levitating magic trick is 70 kg and the mass of the stainless steel rods is 7.8 gm/cm. Centroids of upper and lower square lie on the same vertical line.(You can assume that the center of mass of the body lies on the centroid of the upper square)



a. (7.37,2.45,0)
b. (24.76,34.93,0)
c. (39.5,39.8,0)
d. (40,40,0)

3. In this magic trick, if you rearrange the components of the bigger triangle, you will get an empty square tile. What do you think is the reason behind this?





a. The rearranging of the component shapes causes a change in area resulting in an open square tile.
b. The angle of the hypotenuse of the bigger triangle is different in the initial and the final case causing an open square tile.
c. There is a gap between two shapes in the initial figure so, a block would remain open always.
d. The figure in the initial condition is not a triangle, so the empty square block remains.

4. Why do you think there is a hole in the outer vessel in the infinite water magic trick?
a. It increases pressure in the outer vessel when closed by a finger, causing water to enter through the inner hole.
b. When we release our finger from the outer hole, the water enters the glass through the inner hole due to level difference.
c. When we close the hole with our finger, it prevents water from coming back in the outer vessel through the inner hole.
d. We can add water through a pipe from the outer hole.

Q5 and Q6 are interlinked. You can assume that for both the ambient temperature is 27°C and atmospheric pressure as 1 atm.

5. In the magic trick where the balloon is deflated what could be the temperature of the liquid that is being poured on the balloon?

a. 200°C
b. -200°C
c. 100°C
d. 30°C

6. What would be the deflated volume of helium gas present in the balloon if the initial diameter of the spherical balloon was 50cm.

a. 15.93 L
b. 0 L
c. 0.16 L
d. 5.264 L

7. What would be the minimum number of points that a magician needs to win to qualify in the knockouts?

a. 30
b. 25
c. 10
d. 15

8. **Which of the following statements are true?

a. The winner of the tournament will have the highest number of wins.
b. There will be 4 magicians in the knockout stage having only 1 win.
c. The winner may have the same number of wins as a magician eliminated in the qualifying stage.
d. A magician may win more than half the matches in the tournament but still may not qualify